

Spatial-Temporal Multi-Cue Network for Sign Language Recognition and Translation

Hao Zhou , Wengang Zhou , Yun Zhou, and Houqiang Li , *Fellow, IEEE*

Abstract—Despite the recent success of deep learning in video-related tasks, deep models typically focus on the most discriminative features, ignoring other potentially non-trivial and informative contents. Such characteristic heavily constrains their capability to learn implicit visual grammars in sign videos behind the collaboration of different visual cues (i.e., hand shape, facial expression and body posture). To this end, we approach video-based sign language understanding with multi-cue learning and propose a spatial-temporal multi-cue (STMC) network to solve the vision-based sequence learning problem. Our STMC network consists of a spatial multi-cue (SMC) module and a temporal multi-cue (TMC) module. The SMC module learns to spatial representation of different cues with a self-contained pose estimation branch. The TMC module models temporal corrections from intra-cue and inter-cue perspectives to explore the collaboration of multiple cues. A joint optimization strategy and a segmented attention mechanism are designed to make the best of multi-cue sources for SL recognition and translation. To validate the effectiveness, we perform experiments on three large-scale sign language benchmarks: PHOENIX-2014, CSL and PHOENIX-2014-T. Experimental results demonstrate that the proposed method achieves new state-of-the-art performance on all three benchmarks.

Index Terms—Multi-cue, pose estimation, segmented attention, sign language recognition, sign language translation.

I. INTRODUCTION

SIGN language is a visual language which serves as the primary communication means of the deaf community. To facilitate the daily communication between the deaf-mute and the hearing people, it is significant to develop sign language translation techniques. Recently, sign language (SL) processing research [1], [2] has gained considerable attention [3]–[8] in the research community. In this paper, we concentrate on sign language (SL) recognition and translation. SL recognition aims to



Fig. 1. The difference between SL recognition and translation tasks. The glosses predicted by SL recognition is temporally consistent with the sign video. The spoken language sentence generated by SL translation is more clear and coherent but has a different word order with the former.

recognize a series of continuous signs to its corresponding sign glosses, while SL translation intends to further generate natural spoken language translations [9]. In brief, SL recognition concentrates on the learning of video representation as a pre-task, while SL translation cares more about the quality of linguistic expression (See Fig. 1).

Sign language involves the collaboration of multi-cues beyond hand gesture. To effectively and accurately express the desired idea, sign language simultaneously leverages both manual elements from hands and non-manual elements from the face and upper-body posture [10]. To be specific, manual elements include the shape, position, orientation and movement of both hands, while non-manual elements include the eye gaze, mouth shape, facial expression and body pose. Human visual perception allows us to process and analyze these simultaneous yet complex information without much effort. However, with no expert knowledge, it is difficult for a deep neural network to discover the implicit collaboration of multiple visual cues automatically. Especially for continuous SL recognition, the transitions between sign glosses may come with temporal variations and switches of different cues.

To acquire multi-cue information, some methods resort to external tools. In [4], [11], an off-the-shelf detector is utilized for hand detection, together with a tracker to cope with shape variation and occlusion. Some methods adopt multi-stream networks with inferred labels (i.e., mouth shape labels, hand shape labels) to guide each stream to focus on individual visual cue [5]. Despite their improvement, they mostly suffer two limitations: First, external tools impede the end-to-end learning on the differentiable structure of neural networks, resulting in sub-optimal performance. Second, off-the-shelf tools and multi-stream networks bring repetitive feature extraction of the same region, incurring expensive computational overhead for such a video-based translation task.

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The authors are with the CAS Key Laboratory of Technology in Geospatial Information Processing and Application System, Department of Electronic Engineering and Information Science, University of Science and Technology of China, Hefei 230027, China (e-mail: zhouh156@mail.ustc.edu.cn; zhwg@ustc.edu.cn; zhouyun@ustc.edu.cn; lihq@ustc.edu.cn).

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To temporally integrate multi-cue features, an intuitive idea is to concatenate features and feed them into a temporal fusion module. In action recognition, two-stream fusion shows significant performance improvement by fusing temporal features of RGB and optical flow [12], [13]. Nevertheless, the aforementioned fusion approaches are based on two counterpart features in terms of representation capability. But when it turns to multiple diverse cues with unequal feature importance, how to fully exploit the synergy between strong features and weak features is still a non-trivial issue. Moreover, for deep learning based methods, neural networks tend to merely focus on strong features for quick convergence, potentially omitting other informative cues, which limits further performance improvement.

To address the above difficulties, we propose a novel spatial-temporal multi-cue (STMC) framework for SL processing. In the SMC module, we attach two extra deconvolutional layers [14], [15] for pose estimation on the top layer of our backbone. A soft-argmax trick [16] is utilized to regress the positions of keypoints and make it differentiable for subsequent operations in the temporal part. The spatial representations of other cues are acquired by the reuse of feature maps from the intermediate layer. Based on the learned spatial representations, we decompose the temporal modelling part into the intra-cue and inter-cue paths in the TMC module. The inter-cue path fuses the temporal correlations between different cues with temporal convolutional (TCOV) layers. The intra-cue path models the internal temporal dependency of each cue and feeds them to the inter-cue path at different time scales.

To fully exploit the potential on SL recognition, we introduce a joint optimization strategy with connectionist temporal classification (CTC) [17] and keypoint regression, making the whole STMC network end-to-end trainable. For SL translation, a segmented attention (SA) mechanism is designed to provide multi-cue information for the encoder-decoder structure. Extensive experiments demonstrate that our STMC network surpasses previous state-of-the-art models for both SL recognition and translation on three publicly available benchmarks.

This work is an extension of the conference paper [18]. A new segmented attention method is proposed to extract context information from multi-cue sources based on the attention-based SL translation framework. Along with the design concept of multi-cue learning, we present a compact framework with self-contained modules and corresponding optimization strategies towards multi-cue SL recognition and translation.

The remainder of this paper is organized as follows. Section II reviews related works about sign language research. In Section III, we introduce the formulation of our proposed network and its optimization strategy. After that, Section IV provides implementation details and presents the experimental results of the proposed method. Finally, we conclude this work in Section V.

II. RELATED WORK

In this section, we briefly review the related work on continuous SL recognition, SL translation and multi-cue fusion.

SL Recognition: A video-based SL recognition system usually consists of two parts: video representation and sequence learning. Early works utilize hand-crafted features like

HOG [19]–[21] for SL recognition. In [21], Wang *et al.* propose a grassmann covariance matrix (GCM) for sign description. Recently, deep learning based methods have been applied to SL recognition for their strong representation capability. 2D convolutional neural networks (2D-CNN) [11], [22] and 3D convolutional neural networks (3D-CNN) [23] are employed for modelling the appearance and motion in sign language videos. In [24], Cui *et al.* propose to combine 2D-CNN with TCOV layers for spatial-temporal representation. In [7], [25], [26], 3D-CNNs are adopted to learn appearance and motion features in sign language videos.

Sequence learning in continuous SL recognition is to learn the correspondence between video sequence and sign gloss sequence. Koller *et al.* [27]–[29] propose to integrate 2D-CNNs with hidden markov models (HMM) to model the state transitions. Wang *et al.* [6] present a multi-temporal belief propagation (MTBP) based segmentation approach for continuous sign spotting. In [4], [30], the attention-based encoder-decoder model [31] is adopted to deal with continuous SL recognition in the way of listen-attend-and-spell (LAS) [32] for speech recognition. In [3], [7], [11], [24], [33], connectionist temporal classification (CTC) [17] algorithm is employed as a cost function, which is able to process unsegmented input data. Based on HMM and CTC, a series of works [3], [24], [26], [28], [34] focus on designing iterative strategies for better feature representation of sign language. Overall, the differentiable loss function of CTC makes the end-to-end optimization of SL recognition models possible and brings competitive performance.

SL Translation: SL translation is different from SL recognition mainly in the aspect of sequence learning. To process the difference in word order and grammar between sign language and spoken language, existing neural SL translation works follow the sequence-to-sequence (seq2seq) frameworks [31], [35], [36]. In [9], Camgöz *et al.* contribute the PHOENIX-2014-T corpus with both sign gloss and spoken language annotation. They propose to use attention-based seq2seq models to learn how to translate from spatiotemporal representations and sign glosses. The input of their Sign-to-Text networks is full frames of SL videos. Ko *et al.* [37] propose an SL translation system based on 2D coordinates of human keypoints. Recently, transformer networks [38] have been widely adopted for neural machine translation (NMT). Camgöz *et al.* [8] apply transformers into sequence learning of sign language. Their work explores the multi-task formulation of SL recognition and translation. Few works have discussed the multi-cue problem in SL translation and kept the design concept of multi-cue learning from SL recognition to translation.

Multi-Cue Fusion: The multiple cues of sign language can be separated into categories of multi-modality and multi-semantic. Early works about multi-modality utilize physical sensors [39], [40] to collect the 3D space information, such as depth and infrared maps [6], [25], [41]. With the development of flow estimation, Cui *et al.* [3] explore the multi-modality fusion of RGB and optical flow which achieves state-of-the-art performance on PHOENIX-2014 database.

In contrast, multi-semantic refers to human body parts with different semantics, which is mainly discussed in this paper. Most related works to date rely on off-the-shelf tools (*e.g.*,

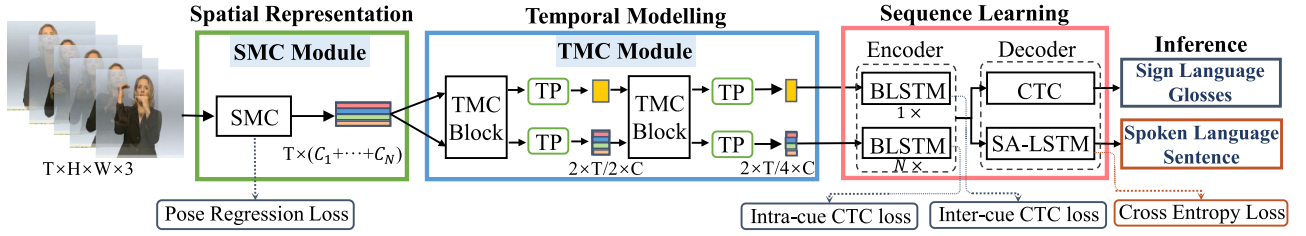


Fig. 2. An overview of the proposed STMC framework. The SMC module is firstly utilized to decompose spatial features of visual cues for T frames in a video. Strips with different colors represent feature sequences of different cues. Then, the feature sequences of cues are fed into the TMC module with stacked TMC blocks and temporal pooling (TP) layers. The output of TMC module consists of feature sequence in the inter-cue path and feature sequences of N cues in the intra-cue path. They are processed by the BLSTM encoder, the CTC decoder and the SA-LSTM decoder for sequence learning and inference. Here, N denotes the number of cues.

pose estimator, facial landmark detector, hand detector and hand tracker) to decompose the multi-cue information from SL videos. Early works use hand-crafted features from segmented hands, tracked body-parts, facial landmarks and trajectories for recognition [10], [20], [42]–[44]. In [4], [11], the feature sequence of hand patches captured by a tracker is early concatenated with the feature sequence of full-frames for SL recognition. Some works resort to weak labels, which lead the neural networks to focus on regions of interest. In [45], Koller *et al.* propose to make use of weak mouth label from spoken German annotation for mouthing recognition. In [5], Koller *et al.* further propose to infer weak hand labels from SL dictionaries. These weak labels are used to establish the state synchronization in HMM of different cues, including full-frame, hand shape and mouth shape. In conclusion, the aforementioned CNN-based works utilize separate backbones for feature extraction of multiple cues and fail to construct spatiotemporal multi-cue interaction within an end-to-end optimized framework.

Different from our previous work [18], we propose a complete framework with multiple customized modules and optimization strategies for multi-cue fusion. Extensive experiments are conducted to show the promising potential of multi-cue framework for both SL recognition and translation.

Learning using Privileged Information (LUPI): The idea of using privileged information is first suggested by V. Vapnik and A. Vashist in [46]. They propose that in a machine learning paradigm, a teacher can provide students with hidden information that exists in explanations, comments and comparisons. The CNN learning-related LUPI methods typically follow the distillation framework [47], where privileged information is uninvolved in inference. In [48], the depth side information is incorporated at the training time of an RGB object detection model through distillation. In [49], Yang *et al.* use one network for the original task and the other one for estimating the loss using privileged information (*e.g.*, bounding boxes as privileged information for multi-object learning). Sharing a similar target of LUPI, our work tries to enable the SL recognition network to have perception ability of the intra-cue information. Differently from the aforementioned methods, although the output of the intra-cue path is not directly used, it is implicitly involved in preceding layers by interaction with the inter-cue path inside the TMC block in inference.

III. PROPOSED APPROACH

In this section, we first introduce the overall architecture of the proposed method. Then we elaborate the key components in our framework, including the spatial multi-cue (SMC) module and temporal multi-cue (TMC) module. Finally, we detail the sequence learning part and the two-stage optimization of our spatial-temporal multi-cue (STMC) framework.

A. Framework Overview

Given a sign video $\mathbf{x} = \{x_t\}_{t=1}^T$ with T frames, the target is to predict its corresponding sign gloss sequence $\mathbf{g} = \{g_u\}_{u=1}^U$ with U glosses and natural spoken language sentence $\mathbf{w} = \{w_v\}_{v=1}^V$ with V words. As illustrated in Fig. 2, our framework consists of three key modules, *i.e.*, spatial representation, temporal modelling and sequence learning. First, each frame is processed by an SMC module to generate spatial features of multiple cues, including full-frame, hand, face and pose. Then, a TMC module is leveraged to capture the temporal correlations of intra-cue features and inter-cue features at different time steps and time scales. Finally, the whole STMC network equipped with the bidirectional Long-Short Term Memory (BLSTM) [50] encoder utilizes CTC and SA mechanism based LSTM (SA-LSTM) decoders for sequence learning and inference.

B. Spatial Multi-Cue Representation

In spatial representation module, 2D-CNN is adopted to generate multi-cue features of full-frame, hands, face and pose. Here, we select VGG-11 model [51] as the backbone network, considering its simple but effective neural architecture design. As depicted in Fig. 3, the operations in SMC are composed of three steps: pose estimation, patch cropping and feature generation.

1) *Pose Estimation:* Deconvolutional networks [14] are widely used in pixel-wise prediction. For pose estimation, two deconvolutional layers are added after the 7th convolutional layer of VGG-11. The stride of each layer is 2. So, the feature maps are $4\times$ upsampled from the resolution 14×14 to 56×56 . The output is fed into a point-wise convolutional layer to generate K predicted heat maps. In each heat map, the position of

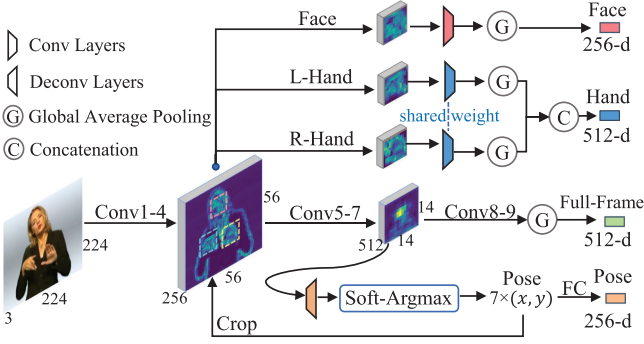


Fig. 3. The SMC Module. The keypoints are estimated for patch cropping of face and hands. The output of SMC includes features from full-frame, hands, face and pose.

its corresponding keypoint is expected to show the highest response value. Here, K is set to 7 for keypoints at the upper body, including the nose, both shoulders, both elbows and both wrists.

To make the keypoint prediction differentiable for subsequent sequence learning, a soft-argmax layer is applied on these heat maps. Denoting K heat maps as $\mathbf{m} = \{m_k\}_{k=1}^K$, each heat map $m_k \in \mathbb{R}^{H \times W}$ is passed through a spatial softmax function as follows,

$$p_{i,j,k} = \frac{e^{m_{i,j,k}}}{\sum_{i=1}^H \sum_{j=1}^W e^{m_{i,j,k}}}, \quad (1)$$

where $m_{i,j,k}$ is the value of heat map m_k at position (i,j) and $p_{i,j,k}$ is the probability of keypoint k at position (i,j) . Afterwards, the expected values of coordinates along x-axis and y-axis over the whole probability map are calculated as follows,

$$(\hat{x}, \hat{y})_k = \left(\sum_{i=1}^H \sum_{j=1}^W \frac{i-1}{H-1} p_{i,j,k}, \sum_{i=1}^H \sum_{j=1}^W \frac{j-1}{W-1} p_{i,j,k} \right). \quad (2)$$

Here, $J_k = (\hat{x}, \hat{y})_k \in [0, 1]$ is the normalized predicted position of keypoint k . The corresponding position of $(\hat{x}, \hat{y})$ in a $H \times W$ feature map is $(\hat{x} \times (H-1) + 1, \hat{y} \times (W-1) + 1)$.

2) *Patch Cropping*: The perception of detailed visual cues is vital, including eye gaze, facial expression, mouth shape, hand shape and orientations of hands. Our model takes predicted positions of the nose and both wrists as the center points of the face and both hands. The patches are cropped from the output ($56 \times 56 \times C_4$) of 4th convolutional layer of VGG-11. The cropping sizes are fixed to 24×24 for both hands and 16×16 for the face. It's large enough to cover body parts of a signer whose upper body is visible to the camera. The center point of each patch is clamped into a range to ensure that the patch would not cross the border of the original feature map.

3) *Feature Generation*: After K keypoints are predicted, they are flattened to a 1D-vector with dimension $2 \sim K$ and passed through two fully-connected (FC) layers with ReLU to get the feature vector of pose cue. Then, feature maps of the face and both hands are cropped and processed by several convolutional layers (see Table I and II), separately. Most sign gestures rely on the cooperation of both hands. So we use weight-sharing convolutional layers for both hands. The outputs of them are

TABLE I
THE HAND BRANCH ARCHITECTURE IN THE SMC MODULE. {LAYER} $\times$ 2
MEANS STACKING TWO LAYERS WITH THE SAME STRUCTURE

Layer	Stride	Kernel	Output Size
hand patches	-	-	$24 \times 24 \times 256$
Conv2D-ReLU	1, 1	3, 3	$24 \times 24 \times 256$
MaxPool2D	2, 2	2, 2	$12 \times 12 \times 256$
{Conv2D-ReLU} $\times$ 2	1, 1	3, 3	$12 \times 12 \times 256$
MaxPooling2D	2, 2	2, 2	$6 \times 6 \times 256$
Conv2D-ReLU	1, 1	3, 3	$6 \times 6 \times 256$
AvgPooling2D	6, 6	6, 6	$1 \times 1 \times 256$

TABLE II
THE FACE BRANCH ARCHITECTURE IN THE SMC MODULE. {LAYER} $\times$ 2
MEANS STACKING TWO LAYERS WITH THE SAME STRUCTURE

Layer	Stride	Kernel	Output Size
face patches	-	-	$16 \times 16 \times 256$
{Conv2D-ReLU} $\times$ 2	1, 1	3, 3	$16 \times 16 \times 256$
MaxPool2D	2, 2	2, 2	$8 \times 8 \times 256$
{Conv2D-ReLU} $\times$ 2	1, 1	3, 3	$8 \times 8 \times 256$
AvgPooling2D	8, 8	8, 8	$1 \times 1 \times 256$

concatenated along the channel-dimension. Finally, we perform global average pooling over the feature maps of full-frame, hands and face to form feature vectors of them (e.g., full-frame: $T \times 7 \times 7 \times 512 \rightarrow T \times 512$).

All features are extracted by passing frames $\mathbf{x} = \{x_t\}_{t=1}^T$ through our spatial multi-cue (SMC) module as follows,

$$\left\{ \{f_{t,n}\}_{n=1}^N, \{J_{t,k}\}_{k=1}^K \right\}_{t=1}^T = \{\Omega_\theta(x_t)\}_{t=1}^T, \quad (3)$$

where $\Omega_\theta(\cdot)$ denotes SMC module and θ denotes the parameters of it. $J_{t,k} \in \mathbb{R}^2$ is the position of keypoint k at the t -th frame. $f_{t,n} \in \mathbb{R}^{C_n}$ is the feature vector of visual cue n at the t -th frame. In this paper, we set $N = 4$, which represents visual cues of full-frame, hand, face and pose, respectively.

C. Temporal Multi-Cue Modelling

Instead of simple fusion, our proposed temporal multi-cue (TMC) module intends to integrate spatiotemporal information from two aspects, intra-cue and inter-cue. The intra-cue path captures the unique features of each visual cue. The inter-cue path learns the combination of fused features from different cues at different time scales. Then, we define a TMC block to model the operations between the two paths as follows,

$$(o_l, f_l) = \text{Block}_l(o_{l-1}, f_{l-1}), \quad (4)$$

where (o_{l-1}, f_{l-1}) and (o_l, f_l) are the input pair and output pair of the l -th block. $o_l \in \mathbb{R}^{T \times C_o}$ denotes the feature matrix of the inter-cue path. $f_l \in \mathbb{R}^{T \times C_f}$ denotes the feature matrix of the intra-cue path which is the concatenation of vectors from different cues along channel-dimension. As the first input pair, $o_1 = f_1 = [f_{1,1}, f_{1,2}, \dots, f_{1,N}]$, where $[\cdot]$ is the concatenating operation and N is the number of cues.

The detailed operations inside each TMC block are shown in Fig. 4 and can be decomposed into two paths as follows. (C is the number of output channels in each path)

1) *Intra-Cue Path*: The first path is to provide unique features of different cues at different time scales. The temporal

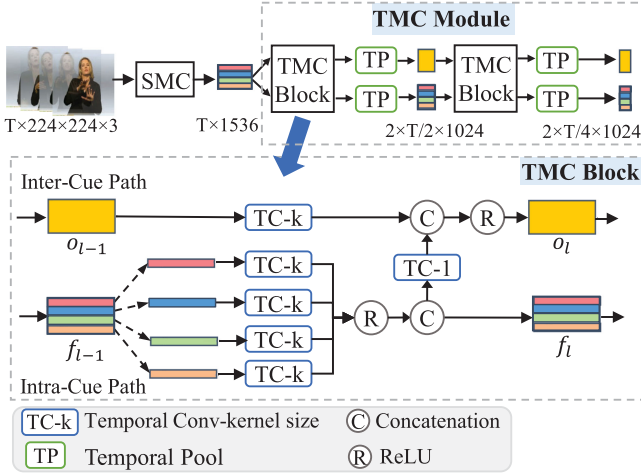


Fig. 4. The TMC Module.

transformation inside each cue is performed as follows,

$$f_{l,n} = \text{ReLU}(\mathcal{K}_k^{\frac{C}{N}}(f_{l-1,n})), \quad (5)$$

$$f_l = [f_{l,1}, f_{l,2}, \dots, f_{l,N}]. \quad (6)$$

Here, $f_{l,n} \in \mathbb{R}^{T \times \frac{C}{N}}$ denotes the feature matrix of n -th cue. $\mathcal{K}_k^{\frac{C}{N}}$ denotes the kernel of a temporal convolution, where k is the temporal kernel size and $\frac{C}{N}$ is the number of output channels.

2) *Inter-Cue Path*: The second path is to perform the temporal transformation on the inter-cue feature from the previous block and fuse information from the intra-cue path as follows,

$$o_l = \text{ReLU}([\mathcal{K}_k^{\frac{C}{2}}(o_{l-1}), \mathcal{K}_1^{\frac{C}{2}}(f_l)]), \quad (7)$$

where $\mathcal{K}_1^{\frac{C}{2}}$ is a point-wise temporal convolution. It serves as a projection matrix between the two paths. Note that f_l is the output of intra-cue path in the present block.

After each block, a temporal max-pooling with stride 2 and kernel size 2 is performed. In this paper, we use two blocks in the TMC module. The kernel size k of all temporal convolutions is set to 5, except the point-wise one. The number of output channels C in each path is set to 1024.

D. Sequence Learning

With the proposed SMC and TMC module, the network can generate inter-cue feature sequence $\mathbf{o} = \{o_t\}_{t=1}^{T'}$ and N intra-cue feature sequences $\mathbf{f}_n = \{f_{t,n}\}_{t=1}^{T'}$. Here, T' is the temporal length of the final output of the TMC module. The question then is how to utilize these two feature sequences to model the conditional probability $p(\mathbf{g}|\mathbf{o}, \mathbf{f})$ and $p(\mathbf{w}|\mathbf{o}, \mathbf{f})$ for the sequence learning.

1) *BLSTM Encoder*: Recurrent neural networks (RNN) can use their internal state to model the state transitions in the sequence of inputs. Here, we use RNN to map the spatial-temporal feature sequence to its sign gloss sequence. RNN takes the feature sequence as input and generates T' hidden states as follows,

$$h_t = \text{RNN}_{\text{en}}(h_{t-1}, o_t), \quad (8)$$

in which h_t is the hidden state at time step t and the initial state h_0 is a fixed all-zero vector. In our approach, we choose the bidirectional Long Short-Term Memory (BLSTM) [50] unit as the recurrent unit for its ability in processing long-term dependencies. BLSTM concatenates forward and backward hidden states from bidirectional inputs. Afterwards, the hidden state of each time step is passed through a fully-connected layer and a softmax layer,

$$q_t = W_e h_t + b_e, \quad y_{t,j} = \frac{e^{q_{t,j}}}{\sum_k e^{q_{t,k}}}, \quad (9)$$

where $y_{t,j}$ is the probability of gloss j at time step t . In SL recognition task, gloss j comes from a given vocabulary.

2) *CTC Decoder for SL Recognition*: Our model employs connectionist temporal classification (CTC) [17] to tackle the problem of mapping video sequence $\mathbf{o} = \{o_t\}_{t=1}^{T'}$ to ordered sign gloss sequence $\mathbf{g} = \{g_u\}_{u=1}^U$ ($U \leq T'$), where the explicit alignment between them is unknown. The objective of CTC is to maximize the posterior probability of the alignments from the source sequence to the target sequence.

CTC creates an extended vocabulary $\mathcal{V}$ with a blank label “-,” where $\mathcal{V} = \mathcal{V}_{\text{origin}} \cup \{-\}$. The blank label represents stillness and transitions which have no precise meaning. Denote the alignment path of the input sequence as $\pi = \{\pi_t\}_{t=1}^{T'}$, where label $\pi_t \in \mathcal{V}$. The probability of alignment path π given the input sequence is defined as follows,

$$p(\pi|\mathbf{o}) = \prod_{t=1}^{T'} p(\pi_t|\mathbf{o}) = \prod_{t=1}^{T'} y_{t,\pi_t}. \quad (10)$$

Define a many-to-one mapping operation $\mathcal{B}$ which removes all blanks and repeated words in the alignment path (e.g., $\mathcal{B}(II - miss - -you) = I, miss, you$). In this way, we calculate the conditional probability of sign gloss sequence $\mathbf{g}$ as the sum of probabilities of all paths that can be mapped to $\mathbf{g}$ via $\mathcal{B}$:

$$p(\mathbf{g}|\mathbf{o}) = \sum_{\pi \in \mathcal{B}^{-1}(\mathbf{g})} p(\pi|\mathbf{o}), \quad (11)$$

where $\mathcal{B}^{-1}(\mathbf{g}) = \{\pi | \mathcal{B}(\pi) = \mathbf{g}\}$ is the inverse operation of $\mathcal{B}$. The formulation of conditional probability $p(\mathbf{g}|\mathbf{f}_n)$ based on the intra-cue part is same as the above.

3) *SA-LSTM Decoder for SL Translation*: Through the BLSTM encoder, a collection of hidden state sequences $\mathcal{H} = (\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_{N+1}) = \{\{h_{n,t}\}_{t=1}^{T'}\}_{n=1}^{N+1}$ is generated. Here, $\mathbf{h}_1 \in \mathbb{R}^{T' \times C_o}$ comes from the inter-cue part and $\mathbf{h}_n \in \mathbb{R}^{T' \times C_f}$ ($n > 1$) comes from the intra-cue part. The decoding process of spoken language translation is formulated as,

$$s_v = \text{RNN}_{\text{de}}(s_{v-1}, w_{v-1}), \quad (12)$$

where s_v is the v -th hidden state of a uni-directional LSTM recurrent unit and w_{v-1} is the word predicted at the previous step. The state s_0 is initialized by the concatenation of $\{h_{n,T'}\}_{n=1}^{N+1}$ with a linear layer. The autoregressive process starts with a word token “bos” indicating the beginning of a sentence and ends until another word token “eos” indicating the end of a sentence is predicted. The question is how to calculate the conditional

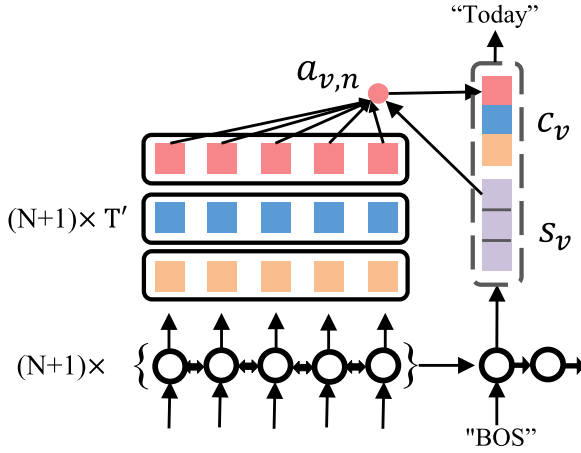


Fig. 5. The illustration of the segmented attention (SA) mechanism. The BLSTM encoders are parallel, including one for inter-cue features and N for intra-cue features. The output of SA-LSTM decoder is segmented along channels to extract the context vector from multi-cue sources.

probability $p(w_v|\mathcal{H})$ at each decoding step by making the most of multi-cue sources.

The idea of attention mechanism is to provide contextual information and make the decoder network focus on a specific range of encoder's outputs. According to Luong's method [36] widely adopted in neural machine translation (NMT), the context vector c_v is calculated as follows,

$$\mathcal{H} = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_{N+1}], \quad (13)$$

$$a_v = \text{softmax}(s_v W \mathcal{H}^T), \quad (14)$$

$$c_v = a_v \mathcal{H}, \quad (15)$$

in which $\mathcal{H} \in \mathbb{R}^{T' \times (C_0 + N C_f)}$ is the concatenation result along the channel axis. Here, attention weights $a_v \in \mathbb{R}^{T'}$ is constrained to be the same for different cues. Hence, it's not appropriate to directly borrow the normal attention (NA) technique for our decoder.

To adapt it to multi-cue framework, we propose a segmented attention (SA) mechanism, depicted in Fig. 5. It separately allocates attention weights to each cue. The context vector c_v is calculated as follows,

$$s_v = [s_{v,1}, s_{v,2}, \dots, s_{v,N+1}], \quad (16)$$

$$a_{v,n} = \text{softmax}(s_{v,n} W_n \mathbf{h}_n^T), \quad (17)$$

$$c_v = [a_{v,1} \mathbf{h}_1, a_{v,2} \mathbf{h}_2, \dots, a_{v,N+1} \mathbf{h}_{N+1}], \quad (18)$$

where s_v is segmented uniformly to $N+1$ parts along channel axis. The probability distribution of words at v -th step is generated with the concatenation of s_v and c_v ,

$$\tilde{s}_v = \tanh(W_s [s_v, c_v] + b_s), \quad (19)$$

$$z_v = \text{softmax}(W_d \tilde{s}_v + b_d). \quad (20)$$

By generating translations word by word, the conditional probability of the sentence $\mathbf{w}$ is formulated as follows,

$$p(\mathbf{w}|\mathbf{o}, \mathbf{f}) = \prod_{v=1}^V p(w_v | w_{1:v-1}, \mathcal{H}) = \prod_{v=1}^V z_{v,w_v} \quad (21)$$

E. Optimization and Inference

The optimization of SL recognition and translation is a progressive process. We adopt a two-stage optimization strategy to demonstrate the effectiveness of STMC on both tasks.

1) *SL Recognition (Stage One)*: With our CTC decoder, the losses of inter-cue feature sequence $\mathbf{o}$ and intra-cue feature sequence $\mathbf{f}_n$ are defined as follows,

$$\mathcal{L}_{\text{CTC-o}} = -\ln p(\mathbf{g}|\mathbf{o}), \quad (22)$$

$$\mathcal{L}_{\text{CTC-f}_n} = -\ln p(\mathbf{g}|\mathbf{f}_n). \quad (23)$$

During the training process, we take the optimization of the inter-cue path as the primary objective. To provide the information of each individual cue for fusion, the intra-cue path plays an auxiliary role. Hence, the joint optimization loss (JL) for SL recognition is given as follows,

$$\mathcal{L}_R = \mathcal{L}_{\text{CTC-o}} + \alpha \sum_n \mathcal{L}_{\text{CTC-f}_n} + \mathcal{L}_P^{[\beta]}. \quad (24)$$

Here, α and β are hyper-parameters, where α is to balance the ratio of auxiliary loss for the intra-cue path, and β is to make the regression loss $\mathcal{L}_P$ of pose estimation have the same order of magnitude with others. Given the estimated keypoints $J_{t,k} \in \mathbb{R}^2$ which is calculated in Eq. 3, its corresponding ground-truth is $\hat{J}$, and the smooth-L1 [52] loss function of pose estimation branch is calculated as follows,

$$\mathcal{L}_P^{[\beta]} = \frac{1}{2TK} \sum_t \sum_k \sum_{i \in (x,y)} \text{smooth}_{L_1} \beta(J_{t,k,i} - \hat{J}_{t,k,i}), \quad (25)$$

in which,

$$\text{smooth}_{L_1}(x) = \begin{cases} 0.5x^2 & \text{if } |x| < 1, \\ |x| - 0.5 & \text{otherwise.} \end{cases} \quad (26)$$

The smooth-L1 loss was proposed in Fast-RCNN [52] to guide the learning of bounding box regression. Compared to L2-norm loss (or mean square error loss), it is more stable in the order of magnitude and more robust when meeting outliers.

For better representation capability, the pseudo-label based pre-training technique [3], [5], [7], [24], [28] is also used for our model before end-end training. The pre-training loss is calculated as follows,

$$\mathcal{L}_{R\text{-pre-train}} = \mathcal{L}_{\text{CE-o}} + \alpha \sum_n \mathcal{L}_{\text{CE-f}_n} + \mathcal{L}_P^{[\beta]}, \quad (27)$$

where CE denotes cross-entropy. See Section IV-B for details.

2) *SL Translation (Stage Two)*: Based on the pre-trained modules from SL recognition, the SA-LSTM decoder are attached to generate spoken language translations. The objective function of SL translation is formulated as follows,

$$\mathcal{L}_T = -\ln p(\mathbf{w}|\mathbf{o}, \mathbf{f}). \quad (28)$$

Here, the function is equivalent to apply cross entropy loss for each word in sentences, according to Eq. 21.

3) *Inference*: For inference, we pass video frames through the SMC and TMC modules. Then, only the inter-cue feature sequence and its BLSTM encoder are used to generate the posterior probability distribution of glosses at all time steps. Finally, words in the spoken language sentence are predicted through the SA-LSTM decoder autoregressively with the outputs of inter-cue and intra-cue encoders. The beam search strategy [53] is adopted to search the most probable sequence within an acceptable range for both tasks.

IV. EXPERIMENTS

A. Dataset and Evaluation

1) *Dataset*: We evaluate our method on three datasets, including PHOENIX-2014 [10], CSL [4], [30] and PHOENIX-2014-T [9]. All three datasets have sign gloss annotations. Only the PHOENIX-2014-T corpus has annotations of spoken language translation.

PHOENIX-2014 is a publicly available German Sign Language dataset, which is the most popular benchmark for continuous SL recognition. The corpus was recorded from broadcast news about weather. It contains videos of 9 different signers with a vocabulary size of 1295. The split of videos for Train, Dev and Test is 5672, 540 and 629, respectively. Our method is evaluated on the multi-signer database.

CSL is a Chinese Sign Language dataset, which has 100 sign language sentences about daily life with 178 words. Each sentence is performed by 50 signers and there are 5000 videos in total. For pre-training, it also provides a matched isolated Chinese Sign Language database, which contains 500 words. Each word is performed 10 times by 50 signers.

PHOENIX-2014-T is an extended version of PHOENIX-2014 and has two-stage annotations for new videos. One is sign gloss annotations for sign recognition task. Another is German translation annotations for SL translation task. The split of videos for Train, Dev and Test is 7096, 519 and 642, respectively. It has no overlap with the previous version between Train, Dev and Test set. The vocabulary size is 1115 for sign gloss and 3000 for German.

2) *Pose Annotation*: To obtain the keypoint positions for training, we use the publicly available HRNet [54] toolbox to estimate the positions of 7 keypoints in upper-body for all frames on three databases. The toolbox gives 2D coordinates (x, y) in the pixel coordinate system. We thus represent each keypoint with a tuple of $(\hat{x}, \hat{y})$ and record it as an array of 7 tuples. The keypoint position (x, y) in an $H \times W$ image is normalized to $(\frac{x-1}{H-1}, \frac{y-1}{W-1})$.

3) *Evaluation*: In SL recognition task, Word Error Rate (WER) is used as the metric of measuring the similarity between two gloss sequences [10]. It measures the least operations of substitution (sub), deletion (del) and insertion (ins) to transform the hypothesis to the reference.

To measure the SL translation performance, BLEU [55] and ROUGE [56] scores are adopted. They are commonly used for NMT. Here, ROUGE refers to ROUGE-L F1-Score [56].

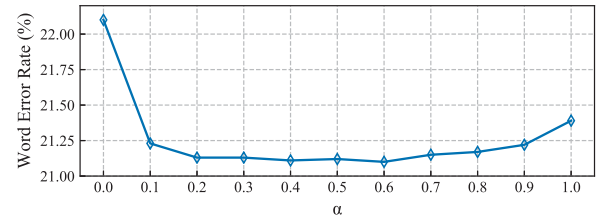


Fig. 6. The effect of weight parameter α in Eq. 24 on PHOENIX-2014 Dev.

B. Implementation Details

In our experiments, the input frames are resized to 224×224 . For data augmentation in one video, we use random shift vertically and horizontally by -15% to 15% (pixels out of borders are set to black), random discard of 20% frames and random flip (50%) of all frames. For inter-cue features, the number of output channels after TCOVs and BLSTM are all set to 1024. There are 4 visual cues. For each intra-cue feature, the number of output channels after TCOVs and BLSTM are all set to 256. In terms of the SA-LSTM decoder, the word embedding size is 512 and the hidden size is 1280 $((4+1) \times 256)$. All structures with LSTM recurrent unit in our model are stacked with 2 layers.

Following the previous methods [3], [7], [24], [28], [34], we adopt an iterative optimization strategy. First, we train a VGG11+1D-CNN+BLSTM network using CTC loss, with full frames as input. We adopt the same temporal 1D-CNN structure as DNF [3] and perform iterative optimization once. The network above with WER(%) 25.66/25.67 on PHOENIX-2014 Dev/Test is used to estimate pseudo label [34] for each clip. Then, we add a fully-connected layer after each output of the last temporal pooling block in the TMC module, including inter-cue feature and intra-cue features. The STMC network without BLSTM is fine-tuned with cross-entropy and smooth-L1 loss by SGD optimizer as in Eq. 27. In each epoch, clips with blank label are sampled within 3 k to control the quantity. The batch size is 24 and the clip size is 16. Finally, with fine-tuned parameters, our STMC network with the CTC decoder is trained end-to-end under joint loss optimization. We use Adam optimizer with learning rate 5×10^{-5} and set the batch size to 2. In all experiments, we set α to 0.6 and β to 30. In fact, the experiment results are insensitive to the slight change of α (see Fig. 6), except $\alpha = 0$. The parameter α and β is to make the numerical value of losses in the same order of magnitude, when these losses have different formulas (e.g., Eq. 22, 25).

The pre-trained backbone model from the recognition stage is shared for SL translation. For training of SA-LSTM, an adam optimizer is adopted with learning rate 5×10^{-5} and batch size 1. Scheduled sampling [57] is a widely adopted technique for RNN-based encoder-decoder models to accelerate the convergence by tuning the teacher forcing [58] rate. It is set to 70% with no decrease during our SL translation training (i.e., 70% inputs of SLT decoder are from ground truths and 30% are from predicted words in the previous step).

Our network architecture is implemented in PyTorch. For fine-tuning, we train the backbone without BLSTM for 25 epochs. Afterwards, the STMC network with the CTC decoder is trained

TABLE III
EVALUATION OF DIFFERENT CUE COMBINATIONS FOR SL TRANSLATION ON PHOENIX-2014-T (THE HIGHER THE BETTER)

	Cue				Dev					Test				
	full	hand	face	pose	ROUGE	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	BLEU-1	BLEU-2	BLEU-3	BLEU-4
0	✓				44.30	44.06	32.69	25.45	20.74	44.70	45.08	33.80	26.44	21.55
1		✓			42.85	41.95	30.88	24.07	19.57	41.93	41.79	30.85	24.07	19.65
2			✓		42.09	41.81	30.31	23.37	18.95	41.91	41.67	30.65	23.69	19.19
3				✓	33.36	32.38	21.88	16.04	12.48	33.08	32.43	22.78	17.10	13.70
4	✓	✓			46.53	46.44	35.25	27.98	22.94	45.44	45.86	34.75	27.43	22.49
5	✓		✓		47.33	47.28	36.06	28.80	23.70	46.30	47.52	36.07	28.56	23.50
6		✓	✓		47.32	46.93	36.11	28.95	23.95	46.57	47.05	36.05	28.66	23.51
7	✓	✓	✓		47.31	46.96	36.09	28.90	24.01	46.56	46.79	35.73	28.50	23.60
8		✓	✓	✓	47.55	47.05	36.06	28.67	23.54	46.71	46.87	35.79	28.34	23.26
9	✓	✓	✓	✓	48.24	47.60	36.43	29.18	24.09	46.65	46.98	36.09	28.70	23.65

TABLE IV
EVALUATION OF ENCODER-DECODER COMBINATIONS FOR SL TRANSLATION ON PHOENIX-2014-T. (NA: NORMAL ATTENTION, SA: SEGMENTED ATTENTION, CSLR: CONTINUOUS SL RECOGNITION)

Method	Pre-trained on CSLR	Dev					Test				
		ROUGE	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	BLEU-1	BLEU-2	BLEU-3	BLEU-4
VGG11+1D-CNN+NA		38.36	39.37	28.46	21.97	17.81	38.71	38.90	28.59	22.10	17.96
SMC+TMC+SA		39.76	40.73	29.42	22.61	18.21	39.82	41.05	29.92	23.01	18.47
VGG11+1D-CNN+NA	✓	44.54	44.05	33.44	26.36	21.71	44.19	44.45	33.55	26.27	21.42
SMC+1D-CNN+NA	✓	45.03	45.30	33.94	26.74	21.76	44.68	46.06	34.85	27.51	22.31
SMC+TMC+NA	✓	46.57	45.81	34.73	27.69	23.01	45.99	45.88	34.76	27.42	22.43
SMC+TMC+SA	✓	48.24	47.60	36.43	29.18	24.09	46.65	46.98	36.09	28.70	23.65

TABLE V
EVALUATION OF DIFFERENT MODULE COMBINATIONS FOR SL RECOGNITION ON PHOENIX-2014 (THE LOWER THE BETTER). (JL: JOINT OPTIMIZATION LOSS)

Method	Dev		Test	
	del/ins	WER	del/ins	WER
VGG+1D-CNN	9.0/3.0	25.1	8.4/3.0	25.5
SMC+1D-CNN	7.6/3.8	22.7	7.4/3.5	22.4
SMC+1D-CNNs w/ JL	6.6/4.3	21.7	6.7/3.7	21.8
SMC+TMC	6.6/4.7	22.1	6.7/3.8	21.9
SMC+TMC w/ JL	7.7/3.4	21.1	7.4/2.6	20.7

TABLE VI
EVALUATION OF DIFFERENT PATHS IN THE TMC MODULE FOR SL RECOGNITION ON PHOENIX-2014

Method	Dev		Test	
	del/ins	WER	del/ins	WER
Full-Frame	10.5/3.0	24.6	9.8/2.8	24.6
Hand	10.2/3.9	30.5	10.0/3.7	30.7
Face	14.1/4.8	34.9	13.7/4.6	34.7
Pose	20.6/3.2	51.4	18.8/3.3	49.5
Inter-cue	7.7/3.4	21.1	7.4/2.6	20.7

end-to-end for 30 epochs. Finally, the BLSTM encoder with the SA-LSTM decoder is trained for 30 epochs. We recorded the training time on PHOENIX-2014-T with 7096 samples. Each epoch in SL pre-training takes 0.78 hours. Each epoch in SL recognition training takes 0.91 hours. Each epoch in SL translation training takes 0.30 hours. For inference, the beam width is set to 20 for recognition and 3 for translation. Experiments are run on 4 RTX 1080Ti GPUs.

C. Framework Effectiveness Study

For a fair comparison, experiments in this subsection are conducted on PHOENIX-2014 and PHOENIX-2014-T, which are the most popular datasets in SL tasks.

1) *Module Analysis in SL Recognition*: We analyze the effectiveness of each module in our proposed approach. In Table V, different combinations of spatial and temporal modules are evaluated. The baseline model is composed of VGG11 and 1D-CNN with a BLSTM encoder. With the aid of multi-cue features, the SMC module provides about 3% points improvement compared with baseline on the test set. However, with no extra guidance, the TMC module doesn't show expected gain by replacing the 1D-CNN. With joint loss (JL) optimization, the intra-cue path is

guided by CTC loss to learn temporal dependency of each cue and provides 1.2% points extra gain on the test set. Compared with the baseline model, our STMC network reduces the WER on the test set by 4.8% points.

2) *Pre-Training and Module Analysis in SL Translation*: The multi-task training and structural changes both account for the increased performance of our models. In Table IV, we present the impact of continuous SL recognition (CSLR) pre-training on PHOENIX-2014-T. It brings 5.18 BLEU-4 score boost for SMC+TMC+SA, which shows that the SL translation models largely benefit from CSLR pre-training in stage one.

To verify the effectiveness of multi-cue modules and SA mechanism, different combinations of encoder-decoder models are also evaluated in Table IV. With the equipment of the SMC and TMC modules, the translation performance shows 1.8 and 1.0 score improvement of ROUGE and BLEU-4 on test set. Compared with directly using NA mechanism [36], our proposed SA mechanism achieves further performance boost across all translation metrics.

3) *Intra-Cue and Inter-Cue Paths*: With further optimization, the BLSTM encoder of each cue in the intra-cue path can also serve as an individual sequence predictor. In Table VI, WERs of different encoders in both paths are evaluated on the

TABLE VII

EVALUATION OF INFERENCE TIME PER VIDEO AND WER PERFORMANCE WITH DIFFERENT BACKBONES ON PHOENIX-2014. (PE: AN EXTERNAL VGG11-BASED MODEL FOR POSE ESTIMATION)

Method	FLOPS	Time	Dev	Test
VGG+1D-CNN	7.5G	264ms	25.1	25.5
SMC w/o deconv+TMC+PE	17.3G	628ms	21.7	21.4
SMC w/ deconv+TMC	10.3G	352ms	21.1	20.7
GoogLeNet (2-stream) [5]	3.1G	524ms	26.0	26.0

dev set. Among the four cues, the performance of pose is worst. With only the position and orientation of joints in upper-body, it's difficult to distinguish the subtle variations in the appearance of sign gestures. The performance of hand is superior to that of face, while full-frame achieves relatively better performance. By leveraging the synergy of different cues, the inter-cue path shows the lowest WER.

Benefited from the flexibility of our SA mechanism, we provide results of different cue combinations for SL translation in Table III. By comparing results of the 4th (full+hand) and 5th (full+face) combinations in Table III, it's clear that facial information is an important element for SL processing. As a structured representation, the pose cue does not bring notable performance boost but still shows effective complement for original full frames.

4) *Parameter Tuning*: The parameter α is to balance the numerical value of losses in joint loss optimization. In Fig. 6, the results of tuning α between $[0.0, 1.0]$ are presented. When $\alpha = 0.0$, it means the joint loss optimization stops working and the error rate increases distinctly comparing with $\alpha \neq 0.0$. It shows that the guidance of single cue ($\mathcal{L}_{CTC-f_n}$) is important for enabling the effectiveness of intra-cue path in the TMC module. When α is controlled in the range of $[0.2, 0.7]$, the performance of SL recognition model is relatively stable.

The selection of β follows the same principle as α . According to experimental observation, the value of CTC Loss is in the range of $[5, 50]$ in the early training stage. For example, if the predicted keypoints (normalized to $[0, 1]$) shift a quarter of input frame size, the value of pose estimation loss should be $0.03 \cdot 125 (= 0.5 \times 0.25^2)$, which is much smaller than CTC loss. To make them in the same magnitude order, we set β to 30 and the value of pose estimation loss becomes 7 ($= |30 \times 0.25| - 0.5$). Actually, the experimental results are not sensitive to slight β change.

5) *Inference Time*: To clarify the effectiveness of the self-contained pose estimation branch, we evaluate the inference time in Table VII. The inference time depends on the video length. In average, it takes around 8 seconds (25FPS) for a sign sentence. For a fair comparison, we evaluate the inference time of 200 frames on a single GPU. Compared with introducing an external VGG-11 based pose estimator, our self-contained branch saves around 44% inference time. With no mouthing labels on PHOENIX-2014, CNN-LSTM-HMM [5] only utilizes 2 GoogLeNets, which still runs more slowly than our network. If we increase the number of cues used in CNN-LSTM-HMM, the inference time of it will multiply increase (the flops of one GPU is limited despite parallel calculation). It's notable that our framework with the self-contained branch shows slightly better performance than an off-the-shelf model. We argue that the

TABLE VIII

QUALITATIVE RESULTS OF SL TRANSLATION. (GT: GROUND TRUTH, NA: NORMAL ATTENTION, SA: SEGMENTED ATTENTION)

GT:	heute nacht neunzehn bis fünfzehn grad im südosten bis zwölf grad. (tonight nineteen to fifteen degrees in the southeast to twelve degrees.)
[8]:	heute nacht werte zwischen neun und fünfzehn grad im südosten bis zwölf grad. (tonight values between nine and fifteen degrees in the southeast to twelve degrees.)
NA:	heute nacht werte zwischen fünfzehn und fünfzehn grad im südosten bis zwölf grad. (tonight values between fifteen and fifteen degrees in the southeast to twelve degrees.)
SA:	heute nacht werte zwischen neunzehn und fünfzehn grad im südosten bis zwölf grad (tonight values between nineteen and fifteen degrees in the southeast to twelve degrees.)
GT:	im süden und südwesten gebietsweise regen sonst recht freundlich. (in the south and southwest partly rain otherwise quite friendly.)
[8]:	im südwesten regnet es zum teil kräftig. (in the southwest it partly rains heavily.)
NA:	in der südwesthälfte regnet es zeitweise im nordosten auch freundlich. (in the southwest half it rains at times in the northeast also friendly.)
SA:	im süden regnet es hier und da sonst auch freundlich. (in the south it rains here and there otherwise also friendly.)
GT:	heute nacht ist es meist stark bewölkt örtlich regnet oder nieselt es etwas. (tonight it is mostly cloudy locally rain or drizzle.)
[8]:	heute nacht ist es verbreitet wolkenverhangen gebietsweise regnet es kräftig. (tonight it is widespread cloudy and partly it rains heavily.)
NA:	heute nacht ist es meist stark bewölkt gebietsweise regnet oder nieselt es. (tonight it is mostly cloudy partly rain or drizzle.)
SA:	heute nacht ist es meist stark bewölkt örtlich regnet oder nieselt es. (tonight it is mostly cloudy locally rain or drizzle.)
GT:	an der saar heute nacht milde sechzehn an der elbe teilweise nur acht grad. (in the saar tonight mild sixteen in the elbe partly only eight degrees.)
[8]:	im rhein und südwesten macht sich morgen nur knapp über null grad. (in the rhine and southwest, tomorrow will be just above zero degrees.)
NA:	an der spree heute nacht milde sechs in der eifel nur kühle acht grad. (in the spree tonight mild six in the eifel only cool eight degrees.)
SA:	in der lausitz um bis sechzehn am oberhein nur kühle acht grad. (in the lausitz at up to sixteen in the upper Rhine only cool eight degrees.)

differentiable pose branch plays the role of regularization and then alleviates the overfitting of neural networks. It drives the backbone to have the perception ability of the upper-body structure instead of only body shapes.

6) *Qualitative Results*: Fig. 7 shows an recognition example generated by different cues. The predicted upper-body keypoints are marked with colored dots and skeletons related are also connected. The result of the inter-cue path can effectively learn correlations of multiple cues and make a better prediction.

In Table VIII, several translation results are presented. The German annotations and their English translations are both provided. Compared with S2(G+T) [8] and NA-LSTM, the generated sentences of our SA-LSTM are more complete in content and more accurate in numbers. Despite a few syntax errors, the main challenge lies in the translation of orientations and named entities. It may be improved by introducing an external knowledge base in the future.

7) *Limitations*: The end-to-end training of the STMC network needs the supervision of keypoints annotations. In our work, the 7 keypoints (i.e., nose, shoulders, elbows, wrists) are estimated by HRNet [54] (~20FPS) and pre-stored locally. With 4 RTX 1080Ti GPUs, the processing of 5000 sign videos takes about 3.5 hours ($= (5000 \times 8 \times 25) / (3600 \times 20 \times 4) = (\#video \times \#second \times \#video-fps) / (3600 \times \#model-fps \times \#GPU)$). It requires more time when pre-preprocessing the dataset than other single-cue methods.

D. Comparison With State-of-The-Art Methods

1) *Evaluation on PHOENIX-2014*: In Table IX, we compare our approach with methods on PHOENIX-2014. CMLLR and 1-Mio-Hands belong to traditional HMM-based model with hand-crafted features. In SubUNets and LS-HAN, full-frame features are fused with features of hand patches, which are captured by an external tracker. In CNN-LSTM-HMM, two-stream

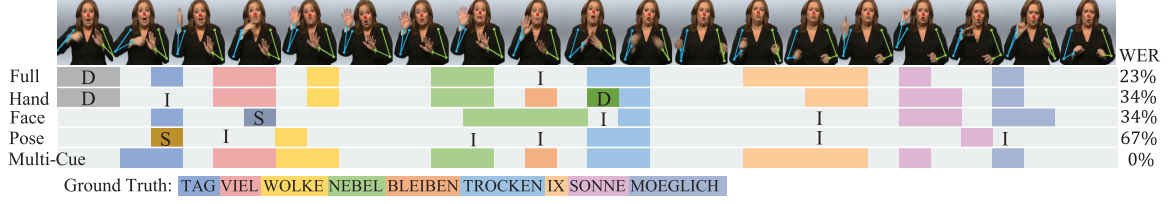


Fig. 7. A qualitative recognition result of different cues with estimated poses (zoom in) from Dev set (D: delete, I: insert, S: substitute).

 TABLE IX
 COMPARISON WITH METHODS FOR SL RECOGNITION ON PHOENIX-2014 (THE LOWER THE BETTER). SMC-R AND STMC-R DENOTE SL RECOGNITION WITH OUR SMC AND STMC MODEL, RESPECTIVELY

Method	Reference	Cue				Dev		Test	
		full	hand	face	pose	del/ins	WER	del/ins	WER
CMLLR [10]	CVIU'15		✓	✓	✓	21.8/3.9	55.0	20.3/4.5	53.0
1-Mio-Hands [27]	CVPR'16		✓	✓	✓	16.3/4.6	47.1	15.2/4.6	45.1
CNN-Hybrid [59]	BMVC'16		✓			12.6/5.1	38.3	11.1/5.7	38.8
SubUNets [11]	ICCV'17	✓	✓			14.6/4.0	40.8	14.3/4.0	40.7
Staged-Opt [24]	CVPR'17		✓			13.7/7.3	39.4	12.2/7.5	38.7
Re-sign [28]	CVPR'17	✓				-	27.1	-	26.8
LS-HAN [4]	AAAI'18	✓	✓			-	-	-	38.3
Hybrid CNN-HMM [29]	IJCV'18		✓			-	31.6	-	32.5
IAN [7]	CVPR'19	✓				12.9/2.6	37.1	13.0/2.5	36.7
CNN-LSTM-HMM [5]	TPAMI'19	✓	✓			-	26.0	-	26.0
DNF [3] (RGB)	TMM'19	✓				7.8/3.5	23.8	7.8/3.4	24.4
DNF [3] (RGB+Flow)	TMM'19	✓				7.3/3.3	23.1	6.7/3.3	22.9
SMC-R (ours)	This work	✓	✓	✓	✓	7.6/3.8	22.7	7.4/3.5	22.4
STMC-R (ours)	This work	✓	✓			6.6/4.0	21.9	6.3/3.6	21.9
STMC-R (ours)	This work	✓	✓	✓	✓	7.7/3.4	21.1	7.4/2.6	20.7

 TABLE X
 COMPARISON WITH METHODS FOR SL TRANSLATION ON PHOENIX-2014-T (TOP: TEXT-TO-TEXT, BOTTOM: VIDEO-TO-TEXT). STMC-T DENOTES SL TRANSLATION WITH OUR STMC MODEL

Method	Pre-trained modules on CSLR task	Dev					Test				
		ROUGE	BLEU-1	BLEU-2	BLEU-3	BLEU-4	ROUGE	BLEU-1	BLEU-2	BLEU-3	BLEU-4
S2G→G2T-RNN [9]	Re-sign [28]	43.76	41.08	29.10	22.16	17.86	43.45	41.54	29.52	22.24	17.79
S2G→G2T-Transformer [8]	CNN-LSTM-HMM [5]	-	47.84	34.65	26.88	21.84	-	47.74	34.37	26.55	21.59
S2G2T-RNN [9]	Re-sign [28]	44.14	42.88	30.30	23.02	18.40	43.80	43.29	30.39	22.82	18.13
S2G2T-Transformer [8]	CNN-LSTM-HMM [5]	-	47.73	34.82	27.11	22.11	-	48.47	35.35	27.57	22.45
S2T-RNN [9]	-	31.80	31.87	19.11	13.16	9.94	31.80	32.24	19.03	12.83	9.58
S2(G+T)-Transformer [8]	CNN-LSTM-HMM [5]	-	47.26	34.40	27.05	22.38	-	46.61	33.73	26.19	21.32
SMC+TMC+Transformer	STMC-R	47.47	46.58	35.09	28.12	23.05	46.03	46.12	35.27	27.52	22.54
STMC-T (ours)	STMC-R	48.24	47.60	36.43	29.18	24.09	46.65	46.98	36.09	28.70	23.65

networks are trained with weak hand labels and sign gloss labels, respectively. The difference between STMC and the above methods lies in the design of module structures. Only CNN-LSTM-HMM and STMC are methods specially designed for the multi-cue problem. Our STMC outperforms two recent multi-cue methods, i.e., LS-HAN and CNN-LSTM-HMM by 17.6% and 5.3% points. Moreover, compared with DNF which explores the fusion of RGB and optical flow modality, STMC still surpasses this best competitor by 2.2% points. Based on the RGB modality, we propose a novel STMC framework and achieves 20.7% WER on the test set, a new state-of-the-art result on PHOENIX-2014.

2) *Evaluation on CSL*: In Table XI, we evaluate our approach on CSL under two settings. CSL dataset contains a smaller vocabulary compared with PHOENIX-2014. Following the works of [4], [30], the dataset is split by two strategies in Table XI. **Split I** is a signer independent test while **Split II** is an unseen sentence test. Between two settings, **Split II** is more challenging for that recognizing unseen combinations of words is difficult. In IAN, their alignment algorithm of CTC decoder and LSTM decoder shows notable improvement,

 TABLE XI
 COMPARISON OF WER WITH METHODS FOR SL RECOGNITION ON CSL. STMC-R DENOTES SL RECOGNITION WITH OUR STMC MODEL

Method	Reference	Cue				Split I	Split II
		full	hand	face	pose		
S2VT [60]	ICCV'15	✓				25.5	67.0
LS-HAN [4]	AAAI'18	✓	✓			17.3	-
HLSTM-attn [30]	AAAI'18	✓				10.2	64.1
CTF [33]	MM'18	✓				11.2	-
DenseTCN [61]	IJCAI'19	✓				14.3	44.7
IAN [7]	CVPR'19	✓				-	32.7
STMC-R (ours)	This work	✓	✓			2.4	29.3
STMC-R (ours)	This work	✓	✓	✓	✓	2.1	28.6

compared with previous methods. Benefiting from multi-cue learning, our STMC framework outperforms the best competitor on CSL by 4.1% points on WER.

3) *Evaluation on PHOENIX-2014-T*: In Table XII, we provide a recognition result of our method on PHOENIX-2014-T. As a newly proposed dataset [9] for SL translation, PHOENIX-2014-T provides an extended database with sign gloss annotation and spoken German annotation. CNN-LSTM-HMM utilizes spoken German annotation to infer the weak

TABLE XII
COMPARISON OF WER WITH METHODS FOR SL RECOGNITION ON
PHOENIX-2014-T. STMC-R DENOTES SL RECOGNITION WITH OUR STMC
MODEL. (F: FULL-FRAME, M: MOUTH, H: HAND, P: POSE)

Method	Reference	Cue				Dev	Test
		full	hand	face	pose		
1 stream [5]	TPAMI'19	✓				24.5	26.5
2 stream [5]	TPAMI'19	✓		✓		24.5	25.4
3 stream [5]	TPAMI'19	✓	✓	✓		22.1	24.1
S2(G+T) [8]	CVPR'20	✓				24.6	24.5
STMC-R (ours)	This work	✓	✓	✓		19.8	21.2
STMC-R (ours)	This work	✓	✓	✓	✓	19.6	21.0

mouth shape labels for each video. It provides results of multi-cue sequential parallelism, including full-frame, hand and mouth. Our method surpasses all three combinations of CNN-LSTM-HMM.

In Table X, we compare our approach with SL translation methods. These two methods are both proposed by Camgöz *et al.*. The RNN-based models [9] only adopt full frame features from Re-sign. The transformer-based [8] models utilize pre-trained features from CNN-LSTM-HMM. These two methods provide results based on the text-to-text setting, which requires an extra decoding step from gloss to translation. Compared with S2(G+T)-Transformer on the video-to-text setting, our model boosts the BLEU-4 metric by 2.3% points. It supports that introducing multi-cue sources also bring notable performance gain for SL translation.

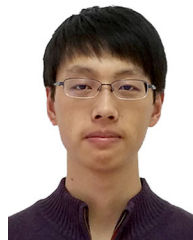
V. CONCLUSION

In this paper, we present a novel multi-cue framework for SL recognition and translation, which aims to learn spatial-temporal correlations of visual cues in an end-to-end fashion. In our framework, a spatial multi-cue module is designed with a self-contained pose estimation branch to decompose spatial multi-cue features. Moreover, we propose a temporal multi-cue module composed of the intra-cue and inter-cue paths, which aims to preserve the uniqueness of each cue and explore the synergy of different cues at the same time. A joint optimization strategy and a segmented attention mechanism are proposed to accomplish multi-cue sequence learning. Extensive experiments on three large-scale sign language datasets demonstrate the superiority of our STMC framework.

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Hao Zhou received the B.S. degree in communication engineering from Xidian University, Xi'an, China, in 2017. He is currently working toward the Ph.D. degree with the Department of Electronic Engineering and Information Science, University of Science and Technology of China, Hefei, China. His research interests include computer vision and sign language processing.



Wengang Zhou received the B.E. degree in electronic information engineering from Wuhan University, Hubei, China, and the Ph.D. degree in electronic engineering and information science from the University of Science and Technology of China (USTC), Hefei, China, in 2006 and 2011, respectively. From 2011 to 2013, he was a Postdoc Researcher with Computer Science Department, University of Texas at San Antonio, San Antonio, TX, USA. He is currently a Professor with the Department of Electrical Engineering and Information Systems, USTC.

His research interests include multimedia information retrieval and computer vision.



Yun Zhou received the B.S. degree in electronic and information engineering from Anhui Normal University, Wuhu, China, in 2010, and the Ph.D. degree in communication and information system from the Hefei University of Technology, Hefei, China, in 2018. She is currently a Postdoctoral Researcher with the Electronic Engineering and Information Science Department, University of Science and Technology of China, Hefei, China. Her research interests include computer vision and reinforcement learning.



Houqiang Li (Fellow, IEEE) received the B.S., M.Eng., and Ph.D. degrees in electronic engineering from the University of Science and Technology of China, Hefei, China, in 1992, 1997, and 2000, respectively. He is currently a Professor with the Department of Electronic Engineering and Information Science, University of Science and Technology of China. He has authored and coauthored more than 200 papers in journals and conferences. His research interests include multimedia search, image/video analysis, video coding, and communication. From 2010 to 2013, he

was an Associate Editor for the IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY. He was the TPC Co-Chair of VCIP 2010 and is the General Co-Chair of ICME 2021. He was the recipient of the National Science Funds (NSFC) for Distinguished Young Scientists, the Distinguished Professor of Changjiang Scholars Program of China, the Leading Scientist of Ten Thousand Talent Program of China, the National Technological Invention Award of China (second class) in 2019, the National Natural Science Award of China (second class) in 2015, the Best Paper Award for VCIP 2012, the Best Paper Award for ICIMCS 2012, and the Best Paper Award for ACM MUM in 2011.